Room: _____

Column: _____

Row: _____

Name and surname: _____

Student ID number: _____

Image-based Biometrics

written exam

2022-01-26

Total points: **100**
Time limit: **60 minutes**
There are 10 questions in this exam. Support your answers with arguments!
Please write legibly and grammatically correct! You can answer in English or Slovene.
Literature is not allowed!
Communication and use of any type of electronic device is forbidden!

1. Explain all three levels of features for face recognition. (10)
2. We skeletonised the binary image of a fingerprint. How do we now extract the minutiae information (including direction)? Sketch and explain. (10)
3. What kind of pattern types can we observe on rolled fingerprint images? What can we conclude based on the number of singular points on them? (10)
4. Explain alignment by geometric warping. Which method gives better results in a more general alignment context and why? (10)
5. Why is the normalisation of iris images needed in a standard iris recognition pipeline? How do we do it? (10)
6. How can we test our trained model using cross validation? Why do we use cross validation? (10)
7. What is an integral image? Where did we use it? How did we use it? Why did we use it? (10)
8. Which modality out of three in-depth discussed at the course gives most interpretable information to a human for determining the gender of a subject based on extracted features? Explain why? (10)
9. Sketch the taxonomy of fusion levels. Which of them are applicable in a verification scenario? (10)
10. Place the formula (10)

$$D = \left[(F_1 + F_2 + F_3); \frac{F_2}{F_1 + F_3} \right]$$

into the context of the course content and explain it.